



Coordinate Geometry · Worksheet 1

Chapter 3: Coordinate Geometry. Answer all questions. Show your working clearly. Total: 106 marks. Time: 2 hours 10 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are five sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- A rough sketch is often the quickest way to check an answer — you are encouraged to draw one.
- Give answers in exact form (fractions, surds) unless the question says otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Straight lines	20	
B Circles	19	
C Circle geometry: tangents, chords and the semicircle theorem	23	
D Parametric equations: converting to Cartesian form	18	
E Parametric equations: sketching and points on a curve	22	
Roar-level challenge	4	
Total	106	

Section A Straight lines

20 marks

1. Find an equation of the line passing through the points $(-2, 5)$ and $(4, -7)$, giving your answer in the form $y = mx + c$. [3]

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2. The line l has equation $3x - 4y + 8 = 0$. [2]
Find the gradient of l and the coordinates of the point where l crosses the y -axis.

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3. Find an equation of the line that passes through the point $(3, -1)$ and is parallel to the line $2x + 5y = 7$. [3]
Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

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4. The line l_1 passes through the point $(-1, 4)$ and is perpendicular to the line $y = 3x - 2$. [4]
(a) Find an equation of l_1 in the form $ax + by + c = 0$, where a , b and c are integers.
(b) Find the coordinates of the point where l_1 crosses the x -axis.

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5. The points A , B and C have coordinates $(-2, 3)$, $(4, 7)$ and $(6, 4)$ respectively.

(a) Show that AB is perpendicular to BC .

[3]

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(b) Find the exact area of triangle ABC .

[3]

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(c) The point D is such that $ABCD$ is a rectangle.
Find the coordinates of D .

[2]

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(Total for Question 5 is 8 marks)



Section B Circles

19 marks

6. Write down an equation of the circle with centre $(3, -5)$ and radius 4. [2]

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7. A circle has equation $(x + 2)^2 + (y - 7)^2 = 45$. [2]
Write down the coordinates of its centre and find its radius, giving your answer as a simplified surd.

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8. A circle has equation [4]
$$x^2 + y^2 + 8x - 10y + 5 = 0.$$

Find the coordinates of the centre of the circle and the radius of the circle.

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9. The points $P(-1, 2)$ and $Q(7, 8)$ are the ends of a diameter of a circle.
Find an equation of the circle.

[4]

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10. The circle C has equation

$$x^2 + y^2 - 6x + 2y + k = 0,$$

where k is a constant.

- (a) Find the coordinates of the centre of C .

[2]

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- (b) Find the range of possible values of k .

[2]

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- (c) Given that C touches the y -axis, find the value of k .

[3]

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(Total for Question 10 is 7 marks)



Section C Circle geometry: tangents, chords and the semicircle theorem **23 marks**

- 11.** The circle C has equation $(x - 1)^2 + (y + 3)^2 = 25$. **[2]**

Show that the point $P(4, 1)$ lies on C .

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- 12.** Using the circle C and the point $P(4, 1)$ from the previous question, find an equation of the tangent to C at P . **[4]**

Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

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- 13.** The points $A(-3, 2)$ and $B(5, 2)$ are the ends of a diameter of a circle. **[3]**
- By considering gradients, show that the point $C(1, 6)$ lies on the circle.

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14. The line $y = x + 1$ meets the circle $x^2 + y^2 = 13$ at the points A and B .

(a) Find the coordinates of A and B .

[3]

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(b) Find the length of the chord AB , giving your answer as a simplified surd.

[2]

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(Total for Question 14 is 5 marks)



15. The circle C has equation

$$x^2 + y^2 - 4x - 2y - 15 = 0.$$

(a) Find the centre and the exact radius of C .

[3]

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(b) The line with equation $y = 2x + k$, where k is a constant, meets C .
Show that the x -coordinates of the points of intersection satisfy

[3]

$$5x^2 + (4k - 8)x + k^2 - 2k - 15 = 0.$$

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(c) Hence find the two values of k for which the line $y = 2x + k$ is a tangent to C .

[3]

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(Total for Question 15 is 9 marks)

**Section D Parametric equations: converting to Cartesian form**

18 marks

16. A curve has parametric equations

[2]

$$x = t + 2, \quad y = t^2.$$

Find a Cartesian equation of the curve in the form $y = f(x)$.

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17. A curve has parametric equations

[2]

$$x = 3t, \quad y = \frac{6}{t}, \quad t \neq 0.$$

Find a Cartesian equation of the curve.

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18. A curve has parametric equations

[3]

$$x = 4 \cos \theta, \quad y = 4 \sin \theta, \quad 0 \leq \theta < 2\pi.$$

Find a Cartesian equation of the curve and describe the curve fully.

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19. A curve has parametric equations

[4]

$$x = 2 + 3 \cos \theta, \quad y = -1 + 3 \sin \theta, \quad 0 \leq \theta < 2\pi.$$

Show that the curve is a circle, and state its centre and radius.

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20. The curve C has parametric equations

$$x = 4t + 3, \quad y = 4t + 8 + \frac{5}{2t}, \quad t > 0.$$

(a) Show that a Cartesian equation of C is

[5]

$$y = \frac{x^2 + 2x - 5}{x - 3},$$

and state the set of possible values of x .

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(b) The point $P(5, 15)$ lies on C . Find the value of t at P .

[2]

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(Total for Question 20 is 7 marks)

**Section E Parametric equations: sketching and points on a curve**

22 marks

21. A curve has parametric equations $x = t^2$, $y = 2t$.

(a) Complete the table of values.

[2]

t	-2	-1	0	1	2
x					
y					

(b) Sketch the curve for $-2 \leq t \leq 2$, marking with an arrow the direction in which t increases.

[1]

(Total for Question 21 is 3 marks)

22. A curve has parametric equations $x = t + 1$, $y = t^2 - 4$.

[3]

Find the coordinates of the points where the curve meets the x -axis.

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23. A curve C has parametric equations $x = t^2 - 1$, $y = 2t + 3$.

[3]

(a) Find the value of t at the point $(8, 9)$ on C .

(b) Show that the point $(8, -3)$ also lies on C .

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24. The curve C has parametric equations $x = 2t$, $y = t^2 - 3$.

[5]

The line with equation $y = x + 5$ meets C at the points A and B .

Find the coordinates of A and B .

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25. The curve C has parametric equations

$$x = t^2 - 4, \quad y = t(t^2 - 4), \quad -3 \leq t \leq 3.$$

- (a) Find the coordinates of the points where C meets the x -axis. [3]

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- (b) Show that C meets the y -axis only at the origin, and state the values of t at which this happens. [2]

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- (c) Show that a Cartesian equation of C is [3]

$$y^2 = x^2(x + 4).$$

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(Total for Question 25 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

26. The circle C has centre $(3, 4)$ and passes through the origin O . [4]
Find an equation of C , and an equation of the tangent to C at O .

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END OF WORKSHEET

Total: 106 marks